Journal of Beijing Institute of Technology, 2023, Vol. 32, No. 4

Establishment and Optimization of Ablation Surrogate Model for Thermal Protection Material

Weizhen Pan⊠, Bo Gao

Abstract: The temperature response calculation of thermal protection materials, especially ablative thermal protection materials, usually adopts the ablation model, which is complicated in process and requires a large amount of calculation. Especially in the process of optimization calculation and parameter identification, the ablation model needs to be called many times, so it is necessary to construct an ablation surrogate model to improve the computational efficiency under the premise of ensuring the accuracy. In this paper, the Gaussian process model method is used to construct a thermal protection material ablation surrogate model, and the prediction accuracy of the surrogate model is improved through optimization.

Keywords: ablation; surrogate model; thermal protection material

1 Introduction

In the design of the thermal protection system, the thermal protection material plays a crucial role. The ablation thermal protection material absorbs heat through the pyrolysis of the material when heated, and the surface emissivity of the carbon layer produced by the pyrolysis increases, so does the thermal radiation from the surface of the material to the outside world. At the same time, a large amount of pyrolysis gas is generated during the pyrolysis process and enters the boundary layer of the surface through the mass ejection effect, resulting in thermal blockage and further preventing the heat flow from transferring to the internal structure[1]. The heat protection mechanism greatly improves the thermal efficiency of ablation thermal protection materials, and moreover the density of ablative thermal protection materials is generally low, making ablative thermal protection materials widely used in re-entry vehicles and interstellar exploration[2–5].

During the heating process, the thermal protection material gradually undergoes complex physical and chemical reactions. As the temperature increases, pyrolysis reactions gradually occur in the thermal protection material. During the pyrolysis reactions, the pyrolysis phase of the material (usually the resin matrix) gradually pyrolyzes to generate pyrolysis gas and carbon layer, and its own mass decreases. The pyrolysis gas enters the boundary layer through diffusion and convection through the internal pores of the material. Secondary reactions also occur as the temperature increases during the transport process[6–8].

In scientific and engineering practice, to deal with such problems with high computational intensity and unknown function form, the common strategy is model approximation technology or surrogate model technology. The surrogate model is driven by data and has nothing to do with the physical process. It is an approximation to the response of the physical process model,

Manuscript received Nov. 29, 2022; revised Jan. 3, 2023; accepted Apr. 27, 2023. The associate editor coordinating the review of this manuscript was Dr. Junsheng Wang. This work was supported by Independent Research and Development Project of CASC (YF-ZZYF-2022-132).

Weizhen Pan is with Beijing Institute of Spacecraft System Engineering, Beijing 100094, China.

Bo Gao is with School of Astronautics, Harbin Institute of Technology, Harbin 150001, China.

⊠ Corresponding author. Email: hitpwz@126.com

DOI: 10.15918/j.jbit1004-0579.2022.141

with a small amount of computation, which can save a lot of computational cost. The establishment of the surrogate model is mainly divided into two parts. The first step is the experimental design. The parameter sample set $\boldsymbol{X}$ is extracted by a certain sampling method from the parameter space, and the output sample set $\boldsymbol{Y}$ is obtained by experimental methods or simulation analysis of $\boldsymbol{X}$, resulting in the samples $[\boldsymbol{X}, \boldsymbol{Y}]$ required for modeling. The second step is to establish a surrogate model by the selected surrogate model construction method. The classic surrogate model technology includes four basic steps: determining the sampling method, selecting the model that fits the sampling points, selecting the fitting method, and verifying the fitting model. The four steps are related to each other. The key issue of surrogate model techniques is to use existing information to build approximation models with sufficient accuracy and minimal effort [9].

Research on experimental design has been carried out for decades, and there are reviews of experimental design in the literature. Wang and Shan [10] summarized and compared different sampling methods and modeling methods, including traditional experimental design methods, such as complete and partial factorial design, central composite design, Box-Behnken and other methods, which tend to sample near the boundary of the design space, and there are fewer samples in the central area; the computer experimental design mainly involves systematic errors rather than random error of physical experiments, so the experimental design should tend to fill the design space rather than concentrate on the boundaries, making traditional methods such as central composite design fail or not as suitable as space filling design methods include simple grid design, orthogonal design, Latin hypercube design, Hammersley sequential design, uniform design, etc. Ruichen Jin et al. [11] confirmed the researchers' consistent recognition of space filling in deterministic computer experimental designs. Sushant et al. [12] summarized the experimental design method, and summarized the evaluation indicators of space filling performance, mainly based on uniformity and distance, including star deviation [13], L2 deviation [14], potential energy criterion , maximum-minimum distance, minimum-maximum distance [15], etc., different sampling methods can be optimized by using space filling performance evaluation indicators, thereby improving their filling performance. Researchers have compared different experimental design methods, showing that Latin hypercube sampling has obvious advantages [16] and Hammersley sampling performs better in large sample sizes [17]; some improvements have been proposed for the Latin hypercube design through the sequential realization method [18], simulated annealing algorithm [19] and continuous local enumeration method [20] to improve the orthogonality and uniformity.

According to the method of establishing surrogate models, surrogate models can be divided into two types: parametric models and non-parametric models. Based on these two modeling strategies, a semi-parametric model is also developed. Parametric models require preselecting the underlying functional form with respect to the parameters so that these basis functions can be parameterized, such as polynomial models. Simple parametric models require fewer data points to obtain computationally fast and physically meaningful models, but parametric models are less flexible, and can only achieve better accuracy when the underlying functional form selected is close to the actual situation [21], so it is only applicable if the functional form is known in advance. The functional form is unknown in a nonparametric model, so it cannot be parameterized with any basis functions, such as smoothing splines and kernel regression. Nonparametric methods attempt to fit functions by using sampled data to obtain the form of the model, rather than "forcing" them to a particular class of models. Therefore, the model changes according to the sampled data, to reflect the nature of the underlying function.

The commonly used surrogate model techniques include Kriging model [22–24], radial basis function model [25,26], response surface model [27,28], support vector machine [29– 31] et al. Many researchers compared different surrogate models [11], showing that the Gaussian process model is not limited by the number of samples, experimental design methods and functional forms, and has high flexibility [32], the Gaussian process model and the radial basis function method can be used for different experimental design methods and sample sizes, with higher accuracy [17], the input-output relationship is always better estimated by Gaussian process model [33], and that the prediction root mean square error of the simple unit Gaussian process model is smaller than that of the multivariate model [34,35].

In recent years, surrogate models have also been gradually used in complex physical and chemical reactions. Xu Wu [36] adopted the maximum and minimum Latin hypercube sampling, and reduced the dimension of the sample output data through principal component analysis (PCA), and established a Gaussian process using the Matérn 5/2 kernel function. Inverse uncertainty quantification of the five input scaling factors in the nuclear reaction simulator by Bayesian method; Piyush M. Mehta et al. [37] used the surrogate model technology to establish the re-entry process of the near-Earth vehicle. Approximate models for ablation, evaporation, denudation and other processes; Chen Xin [38] used improved maximum and minimum Latin hypercube sampling to obtain parameter samples, and established aerodynamic parameters through eigen orthogonal decomposition combined with Gaussian process model and radial basis function model respectively. The surrogate model of the thermal computational fluid dynamics (CFD) calculation model is used to predict the boundary of the hypersonic thermal environment.

It can be seen that for complex computing models with high computational intensity, establishing surrogate models is an effective means to reduce the amount of computation and improve the computational efficiency while obtaining approximate computational accuracy. The commonly used experimental design methods for establishing surrogate models include complete/partial factorial design, central composite design, orthogonal array design, uniform design, Latin hypercube design and various optimization methods. Commonly used surrogate models include polynomial regression model, response surface model, radial basis function model, Gaussian process model (kriging model), etc. Among them, Gaussian process model is widely used in approximate model establishment and parameter identification due to its strong flexibility and high accuracy.

This paper focus on the establishment and optimization of ablation surrogate model for the use in optimization calculation and parameter identification of ablative thermal protection materials. To this end, the Latin hypercube sampling method is used to extract samples of parameters, which is then used as input into the ablation model to calculate the temperature results. The dimension of temperature results is reduced by PCA to obtain the principal component. The input parameters and output results and then utilized as training samples to construct the ablation surrogate model by Gaussian process model method, and the prediction accuracy of the surrogate model is improved through optimization. The flow chart is shown in Fig. 1.

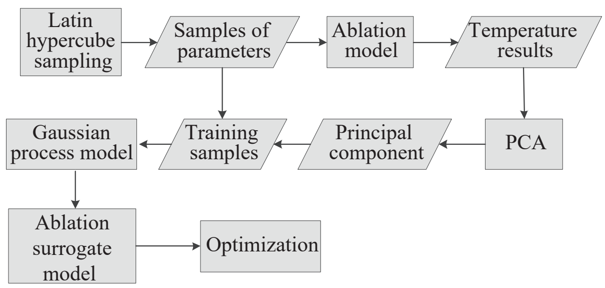


Fig. 1 The flow chart of the research

2 Ablation Model of Thermal Protection Material

The purpose of studying ablative model of materials is to predict the transient thermal response of materials under high heat environment, evaluate the thermal protection property of materials, and guide the design of thermal protection materials and structures.

Under the environment of high temperature, as the material surface is heated, the material temperature gradually raises, thus raw material or one or more components of raw material began to pyrolysis producing pyrolysis gas, which overflows from pyrolysis area to the surface, remaining porous residue (mostly carbon layer). In this process, more complex phenomena will occur, such as the reaction between pyrolysis gas and the carbon layer during the process of overflow, resulting in corrosion or redeposition of the carbon layer. Under the action of external force or thermal stress, the carbon layer is denudated or ruptured, and the enhanced phase leads to melting and mechanical failure, leading to material recession [6,39,40].

Therefore, in solving the internal thermal response of pyrolytic materials, the primary task is to characterize heat transfer and pyrolysis reaction. In the solution process, it is assumed that the model is a one-dimensional heat transfer process, and the heat exchange between the generated pyrolysis gas and the carbon layer is completed instantaneously to reach a heat equilibrium state, and no chemical reaction occurs between the pyrolysis gas and the carbon layer. There is therefore no carbon deposition or chemical denudation. It is also assumed that the resulting pyrolytic gas immediately passes through the carbon layer. By analyzing the material heated in the process of internal energy conservation, chemical reaction process, the internal energy balance of the conservation of mass and material boundary problem, can obtain material internal temperature measuring point, Aero Thermo Company charring material ablation (CMA) [41] of the method is put forward by solving equations and internal energy balance, the decomposition ablative material temperature at the surface energy balance condition, solving process for explicit process, calculation process and related time step and space step, the calculation accuracy is hard to estimate, if the pyrolysis gas rate or ablation back rate faster, calculation easy convergence; NASA Langley Research Center proposed (FIAT) [42] algorithm to solve the thermal response of ablative materials completely implicitly, which reduced the instability in the calculation process of CMA method. The analytical process of ablation model calculation in this paper includes chemical reaction process, internal mass balance equation, internal energy balance equation and surface energy balance equation.

2.1 Chemical Reaction Process

When the thermal protected material is heated, pyrolysis happens inside, which can be characterized by multi-component model. Assuming the decomposition of l components occurs during the heating of thermal protection material, the pyrolysis degree of material at any moment, defined as $c=(\rho-\rho_c)/(\rho_v-\rho_c)$, can be obtained by the following equation

$$c=\sum_{i=1}^{n} c_i\zeta_i \tag{1}$$

where ρ, ρ_v, ρ_c are the density of real-time, original material and residual carbon layer; c_i and ζ_i are the pyrolysis degree and the ratio of weight loss to total weight loss of component i, which can be obtained by thermo-gravimetric analysis test. The rate of pyrolysis reaction of any components satisfy the Arrhenius dynamic reaction equation

$$\frac{\mathrm{d}c_i}{\mathrm{d}t}=-A_i(1-c_i)^{n_i}\exp\left(\frac{-E_i}{RT}\right),\ i=1,2,\cdots,l \tag{2}$$

where A_i, E_i, and n_i are the pre-exponential factor, activation energy and order of reaction [43].

2.2 Internal Mass Balance

The flow of pyrolysis gas generated from the internal reaction of the material from the inside to the outside of the material needs to be considered in the internal mass balance. Assuming that the pyrolysis gas flow is a quasi-steady-state one-dimensional flow, the conservation of mass can be expressed as [44]

$$\frac{\partial \dot{m}_g}{\partial x} = \frac{\partial \rho}{\partial t} \tag{3}$$

where $\dot{m}_g$ is the mass flowing rate; the material density ρ is related to the density of solids ρ_s and gases ρ_g, namely,

$$\rho = \phi\rho_g + \rho_s \tag{4}$$

where ϕ is the material porosity, which can be expressed as a function of the material porosity before (ϕ_v) and after pyrolysis (ϕ_c)

$$\phi = \alpha\phi_v + (1-c)\,\phi_c \tag{5}$$

The density of the solid phase material can be expressed as a function of the density of the material before and after pyrolysis

$$\rho_s = c\rho_v + (1-c)\,\rho_c \tag{6}$$

where ρ_g is the density of the pyrolysis gas produced by the pyrolysis of the material in the pores, which can be expressed as

$$\rho_g = \frac{M_g p}{RT} \tag{7}$$

where M_g is the mass of the gas produced in the pyrolysis process reaching the surface of the material by diffusion and entering the boundary layer, and $\boldsymbol{v}_g$ is the volume average flow rate of the gas, which can be obtained according to Darcy's law [45]

$$\boldsymbol{v}_g = -\frac{1}{\mu\phi}\lambda\frac{\partial p}{\partial x} \tag{8}$$

where λ is the gas permeability of the material and μ is the gas viscosity coefficient.

Substituting into Eq. (3) to obtain the internal mass balance equation as follows

$$\frac{\phi}{p}\rho_{\mathrm{g}}\frac{\partial p}{\partial t} - \frac{\phi}{T}\rho_{\mathrm{g}}\frac{\partial T}{\partial t} + \left(\left(\phi_v - \phi_c\right)\rho_{\mathrm{g}} + \rho_v - \rho_c\right)\frac{\partial c}{\partial t} + \frac{\partial\left(\phi\rho_{\mathrm{g}}v_{\mathrm{g}}\right)}{\partial x} = 0 \tag{9}$$

2.3 Internal Energy Balance

The energy transfer of heat protection materials includes conduction heat transfer and mass transfer heat transfer. During the heating process of the material, pyrolysis occurs inside to generate pyrolysis gas. The pyrolysis gas exchanges heat with the solid material during the escape process, and take the heat out of the surface of material, resulting in mass and heat transfer. According to the assumption of thermal equilibrium state between the pyrolysis gas and the solid material, the temperature of the solid and gas phases at any position is the same. The internal energy transfer of thermal protection material can be characterized by the energy conservation equation of the volume element shown as [46]

$$\frac{\partial(\rho h)}{\partial t} + \frac{\partial}{\partial x}\left(-k\frac{\partial T}{\partial x}\right) + \frac{\partial\left(\phi\rho_g h_g v_g\right)}{\partial x} = Q_{ec} \tag{10}$$

The first term on the left side of Eq. (10) is the energy storage rate, where $h = (\rho_s h_s + \phi\rho_g h_g)/\rho$ is the total enthalpy of the material, which can be determined by the solid enthalpy h_s and the enthalpy value of the pyrolysis gas h_g, and the second term is the conduction heat flow, where k is the material thermal conductivity. The third term is the enthalpy change caused by the flow of pyrolysis gas, where Q_{ec} is the enthalpy change caused by the pyrolysis reaction of the material. The enthalpy of the solid phase and pyrolysis gas can be calculated by the following formula

$$h_s = U_s = \int_{T_0}^{T(t)} c_{p,s}\,\mathrm{d}T \tag{11}$$

$$h_g = U_g + \frac{p}{\rho_g} = \int_{T_0}^{T(t)} c_{v,g}\,\mathrm{d}T + R_g T \tag{12}$$

where $c_{p,s}$ and $c_{v,g}$ are the specific heat of the solid material and the specific heat of the pyrolysis gas at a constant volume.

Substituting Eqs. (11) and (12) into Eq. (10), and simplifying it to

$$
\begin{aligned}
&\left(\rho c_{\mathrm{v}}+\phi \frac{p}{T}-\frac{\phi}{T} \rho_{\mathrm{g}} h_{\mathrm{g}}\right) \frac{\partial T}{\partial t}+ \\
&\left(\left(\rho_v-\rho_c\right) h_{\mathrm{s}}+\left(\phi_v-\phi_c\right) \rho_{\mathrm{g}} h_{\mathrm{g}}\right) \frac{\mathrm{d} c}{\mathrm{~d} t}+ \\
&\frac{\phi}{p} \rho_{\mathrm{g}} h_{\mathrm{g}} \frac{\partial p}{\partial t}+\frac{\partial}{\partial x}\left(-k \frac{\partial T}{\partial x}\right)+h_g \frac{\partial\left(\phi \rho_g \mathrm{v}_g\right)}{\partial x}+ \\
&\phi \rho_g\left(c_{v, g}+\frac{R}{M_{\mathrm{g}}}\right) \frac{\partial T}{\partial x} \cdot \boldsymbol{v}_g=Q_{e c}
\end{aligned} \tag{13}
$$

where $c_v=\left(\rho_s c_{p, s}+\phi \rho_g c_{v, g}\right) / \rho$ is the macroscopic specific heat of the material. The specific heat and thermal conductivity of solid materials are the input quantities of the material database, and both the raw material and the carbonized material are temperature-dependent functions. The specific heat at any position in the pyrolyzing zone can be obtained by linear interpolation of the specific heat of the original material and the carbon layer material using the pyrolysis degree.

$$
c_{p, s}=c c_{p v}+(1-c) c_{p c} \tag{14}
$$

where c_{pv} and c_{pc} are respectively the specific heat of the material before and after pyrolysis, and ρ_v and ρ_c are respectively the density of the material before and after pyrolysis.

Simplify Eqs. (9) × h_g− (13) to get the energy balance equation

$$
\begin{aligned}
&\left(\rho c_{\mathrm{v}}+\phi \frac{p}{T}\right) \frac{\partial T}{\partial t}+\frac{\partial}{\partial x}\left(-k \frac{\partial T}{\partial x}\right)+ \\
&\phi\left(\rho_{\mathrm{g}} c_{v, \mathrm{g}}+\frac{p}{T}\right) \frac{\partial T}{\partial x} \cdot \boldsymbol{v}_{\mathrm{g}}= \\
&Q_{\mathrm{ec}}-\left(\rho_v-\rho_c\right)\left(h_{\mathrm{s}}-h_{\mathrm{g}}\right) \frac{\mathrm{d} \alpha}{\mathrm{d} t}
\end{aligned} \tag{15}
$$

2.4 Surface Energy Balance

Surface boundary conditions usually include the back wall boundary and the heated surface boundary. The back wall boundary conditions are usually designated as known temperature history or convection boundaries. In the design of heat-resistant structures, adiabatic boundary conditions are usually selected as the back wall boundary conditions. There are usually many options for the boundary conditions of the heated surface of the ablated material, including known temperature history or incident convective or radiant heat flow on the non-ablated surface. For ablated surfaces, a more complex surface energy balance is usually used, in which surface chemical reactions during the ablation process should be considered. The surface energy balance can be expressed as [46]

$$
q_0=\varepsilon \sigma\left(T_{\mathrm{surf}}^4-T_{\mathrm{amb}}^4\right)+q_{h w}+\psi\left(q_0-q_{\mathrm{hw}}\right)-\left(k \frac{\partial T}{\partial x}\right) \tag{16}
$$

The left side of the equation is the cold wall heat flux; the first term on the right side is the radiation heat flux q_{rad} from the surface, where ε is the emissivity, T_{surf} and T_{amb} are respectively the temperature of the wall and the ambient; and the second term is the heat loss due to the hot wall effect $q_{\mathrm{hw}}=q_0\left(h_{\mathrm{surf}}-h_0\right) /\left(h_{\mathrm{re}}-h_0\right)$; the third term is the heat loss caused by the thermal blocking effect q_ψ, where ψ is the thermal blocking factor, which can be determined by the following formula

$$
\psi=\beta\left(\frac{M_{\mathrm{a}}}{M_{\mathrm{g}}}\right)^\eta \dot{m}_v \frac{h_{\mathrm{re}}}{q_0} \tag{17}
$$

Among them, the parameters $\beta$and η are determined by the flow state of the boundary layer. In laminar flow, $\beta=0.62$, $\eta=0.26$, in turbulence flow, $\beta=0.20$, $\eta=0.33$ [47]. M_a and $M_g$are respectively the average relative molar masses of boundary layer gas and pyrolysis gas, and $\dot{m}_v$ are the mass loss rate of pyrolysis gas. The fourth item is conduction heat flux q_{cond}. The surface energy balance relationship can be represented by Fig. 2. The control volume shown in Fig. 2 is fixed on the surface, where the energy leaving the control volume includes heat conduction to the inside of the material, heat radiation to the outside of the material, heat loss caused by the hot wall effect, and heat loss by the thermal blocking effect, the heat flowing into the control volume is the cold wall heat flow, including radiation and convective heat flux from the boundary layer.

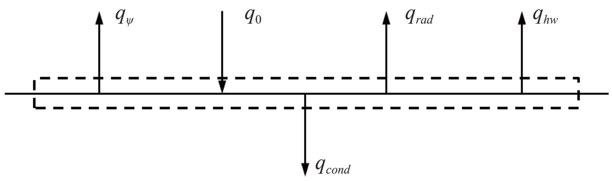


Fig. 2 Surface energy balance diagram of material

3 Gauss Process Model

The Kriging model is based on a statistical process to establish a surrogate model. It was first proposed by Krige, and then applied to computer experimental design and analysis [48] and gradually promoted. It is also known as the Gaussian process model in the field of machine learning. The general form of the Gaussian process model is [49]

$$\hat{f}(x) = \sum_{i=1}^{n_c} w_i \psi_i(x) + \varepsilon(x) \tag{18}$$

Among them, the known basis function $\psi_i(x)$ determines the mean value of the predicted value at sample$x$; $\varepsilon(x)$ is a normal distributed random error with a mean value of zero at sample$x$. The prediction expression of the Gaussian process model is

$$\hat{f}(x) = \psi(x)^{\mathrm{T}} w^* + r(x)^{\mathrm{T}} \gamma^* \tag{19}$$

Among them, $\psi(x) = [\psi_1(x), \cdots, \psi_m(x)]^T$; w^* is the least square estimation of ; $r(x)$ is the correlation vector of $\varepsilon(x)$ and $\varepsilon(x_i)$; w^* and γ^* can be estimated from sample point data:

$$w^* = (\boldsymbol{\Psi}^{\mathrm{T}} R^{-1} \boldsymbol{\Psi})^{-1} \boldsymbol{\Psi}^{\mathrm{T}} R^{-1} y \tag{20}$$

$$\gamma^* = R^{-1}(y - \boldsymbol{\Psi} w^*) \tag{21}$$

where R is the covariance matrix, the elements of which are the correlation between $\varepsilon(x_i)$ and $\varepsilon(x_j)$; $\boldsymbol{\Psi} = [\psi(x^{(1)}), \cdots, \psi(x^{(n)})]^T$; and y is the output data of the sample set. The variance of the predicted value is estimated as

$$s^2(x) = \hat{\sigma}[1 - r^{\mathrm{T}} R^{-1} r] \tag{22}$$

$$\hat{\sigma}^2 = \frac{(y - \boldsymbol{\Psi}^{\mathrm{T}} w^*)^{\mathrm{T}} R^{-1} (y - \boldsymbol{\Psi}^{\mathrm{T}} w^*)}{n} \tag{23}$$

where R and r is determined by the kernel function. The commonly used kernel functions are shown in Tab. 1, where m_j is the distance between two sample points; θ_j is the hyperparameter; d is the input parameter dimension; and K_{v_j} is the v_j order improved Bessel function, by selecting the appropriate one, the derivative performance of the kernel function can be controlled, for example, $v_j = 3/2$ or $v_j = 5/2$ can respectively guarantee the first and second order differential of the Matern kernel function.

Tab. 1 Common used kernel functions

Name	Equation
Square exponent kernel function	$K_{SE}(m_j) = \exp\left(-\sum_{j=1}^{d} \theta_j \lvert m_j \rvert^2\right)$
Linear kernel function	$K_L(m_j) = \max\left\{0, 1 - \sum_{j=1}^{d} \theta_j \lvert m_j \rvert\right\}$
Neural network kernel function	$K_{NN}(x, x') = \theta_j \arcsin\left(\frac{\sigma_w^2 x^T x' + \sigma_b^2}{\sqrt{\sigma_w^2 x^T x + \sigma_b^2 + 1}\sqrt{\sigma_w^2 x'^T x' + \sigma_b^2 + 1}}\right)$
Matern kernel function	$K_M(m_j) = \prod_{j=1}^{d} \frac{1}{(v_j) 2^{v_j - 1}} (\theta_j \lvert m_j \rvert)^{v_j} K_{v_j}(\theta_j \lvert m_j \rvert)$

The hyper-parameters in the kernel function can be obtained through maximum likelihood (ML) estimation. The commonly used maximum log likelihood estimation is [49]

$$\log \mathrm{ML}(\theta) = -\frac{1}{2}[n \ln(\hat{\sigma}^2) + \ln(\lvert \boldsymbol{R}(\theta) \rvert) + (\boldsymbol{y} - \boldsymbol{F}^T \boldsymbol{w}^*)^{\mathrm{T}} \boldsymbol{R}(\theta)^{-1} (\boldsymbol{y} - \boldsymbol{F}^{\mathrm{T}} \boldsymbol{w}^*)] \tag{24}$$

The maximization process of Eq. (24) is the main part of the calculation time in the Gaussian process model. In order to select a suitable training strategy to reduce the calculation consumption of multi-matrix inversion, a lot of research work has been carried out. However, the parameter estimation process still limits this method to low-dimensional problems, usually around 20. The random error term allows the Gaussian process model to give not only the prediction result of the prediction point, but also the variance value of the prediction result. In addition, the Gaussian process model has good approximation ability for nonlinear problems, so the surrogate model of the ablation process is established based on the Gaussian process model in this paper.

4 Principal Component Analysis

The temperature response data usually contains temperature values at different locations at different points, corresponding to the multivariate output of the surrogate model. Special processing is needed to resolve their high correlation and reduce the dimensionality of the output variables. The dimensionality reduction of the output variables can solve this problem. The dimensionality reduction method usually transfers the high-dimensional data to the low-dimensional space through a certain scaling rotation transformation, so that the data in the low-dimensional space can contain most of the high-dimensional data information. Commonly used data dimensionality reduction methods include linear discriminant analysis (LDA) [50,51], principal component analysis (PCA) [52-54], multi-scale transformation (MDS) [55], kernel principal component analysis (KPCA) [56], active subspace method[57] etc. In this paper, PCA is used to reduce the dimensionality of the output data of the ablation model.

PCA projects high-dimensional data into a low-dimensional subspace (also known as the main subspace), so that the projected data has the largest variance and is irrelevant. Principal component analysis can be realized by eigen decomposing the covariance matrix of the data. However, due to numerical calculations, singular value decomposition (SVD) of the data matrix is usually the preferred method.

The temperature data of the internal measuring points of the material may involve a large number of output data. It is assumed that the output data is p dimension, which leads to high-dimensional problems. It is assumed that the input parameters of N group are selected by the sampling method, Calculate the temperature history of the internal measuring points of the material through the ablation model, and obtain the $p\times N$ output data matrix $\boldsymbol{A}$. Since the rows of the data matrix are correlated, the data correlation can be removed by linear transformation $\boldsymbol{PA}=\boldsymbol{B}$, where $\boldsymbol{P}$ is the transformation matrix of $p\times p$, and $\boldsymbol{B}$ is the resulting irrelevant matrix. The process of obtaining the transformation matrix is called principal component analysis, which can be achieved through the following steps[53].

1) Obtain centralized data: Define u as the row average vector of the data matrix, and obtain the centralized data $\boldsymbol{A}_{\text{centered}}$ by subtracting u from each column in the data matrix $\boldsymbol{A}$;

2) Solve the singular value decomposition of $\boldsymbol{A}_{\text{centered}}$

$$\boldsymbol{A}_{\text{centered}}=\boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{V}^{\mathrm{T}} \tag{25}$$

Among them $\boldsymbol{U}$ is an orthogonal matrix of $p\times p$, the columns of which are the left singular vectors of the matrix; $\boldsymbol{V}$ is an orthogonal matrix of $N\times N$, the columns of which are the right singular vectors of the matrix; $\boldsymbol{\Lambda}$ is a diagonal matrix of $p\times N$, the diagonal elements of which are the singular values of the matrix; the non-zero singular values are the square root of the non-zero eigenvalue of $\boldsymbol{A}_{\text{centered}}\boldsymbol{A}^{\mathrm{T}}_{\text{centered}}$ or $\boldsymbol{A}^{\mathrm{T}}_{\text{centered}}\boldsymbol{A}_{\text{centered}}$; the diagonal elements of the matrix $\boldsymbol{\Lambda}$ are arranged in descending order, and the larger singular value corresponds to the important feature in the data matrix $\boldsymbol{A}_{\text{centered}}$.

3) Set $\boldsymbol{P} = \boldsymbol{U}^{\mathrm{T}}$, then:

$$\boldsymbol{P}\boldsymbol{A}_{\text{centered}} = \boldsymbol{U}^{\mathrm{T}}\boldsymbol{A}_{\text{centered}} = \boldsymbol{\Lambda}\boldsymbol{V}^{\mathrm{T}} = \boldsymbol{B} \tag{26}$$

Obviously, the covariance matrix of $\boldsymbol{B}$ is a diagonal matrix. Each row of the matrix $\boldsymbol{P}$ (i.e. each column of the matrix $\boldsymbol{U}$) is called the principal component, or load. The transformed variable (rows of $\boldsymbol{B}$) is called the principal component value, which is the projection of the original data $\boldsymbol{A}_{\text{centered}}$ in the principal component subspace. The eigenvalues of the covariance matrix of $\boldsymbol{A}_{\text{centered}}$ is the main component variance, which corresponds to the square of the diagonal elements of the matrix $\boldsymbol{\Lambda}$.

Determine the subspace dimension p^* of the principal component, which should be less than$p$. Usually the variance of the principal components decreases rapidly, so does the importance of responding principal components. Usually the first p^* principal components are reserved, the principal component variance of which accounts for 95% or 99% of the total variance, forming the $p^* \times p$ transformation matrix $\boldsymbol{P}^*$

$$\boldsymbol{P}^*\boldsymbol{A}_{\text{centered}} = \boldsymbol{B}^* \tag{27}$$

where $\boldsymbol{B}^*$ is the $p^* \times N$ data matrix after dimensionality reduction. So far, the number of output variable is reduced from p to p^*.

After completing the dimensionality reduction of the output data, the principal component value of dimension p^* can be used to replace the p dimension output data in the original space, and $\boldsymbol{B}^*$ be used as a training sample to establish a Gaussian process model. For a given input, the prediction result of the established Gaussian process model should be a $p^* \times 1$ vector $\boldsymbol{b}^*$, where the superscript * represents the prediction result in the principal component subspace, which can be converted to the original space by $\boldsymbol{P}^*$ using the formula

$$\boldsymbol{a} = \boldsymbol{P}^{*\mathrm{T}}\boldsymbol{b}^* + \boldsymbol{u} \tag{28}$$

where $\boldsymbol{a}$ is the $p \times 1$ output vector of the original space changing with time.

5 Ablation Surrogate Model

The above ablation model simplifies the model based on certain assumptions, but the calculation process is still very complicated. In the process of solving inverse ablation problem, the ablation model is usually called repeatedly making the inverse problem solving process is very time-consuming. In order to simplify the model calculation process and improve the efficiency of the inverse problem calculation, it is necessary to establish a surrogate model for the material ablation model. The establishment of an ablation surrogate model includes two steps: experiment design and establishment of the ablation model. Considering that the temperature response data contains temperature values at different measurement points and different moments, the amount of data is usually large, so the principal component analysis is considered to reduce data dimension.

5.1 Experiment Design

In the ablation model, the parameters related to the thermal response of the material include the specific heat capacity, the thermal conductivity and the surface absorption coefficient. The specific heat capacity can be obtained by linear interpolation of the properties of the original material and the carbonized material. In order to facilitate the construction of a surrogate model, it is assumed that the thermal conductivity and surface absorption coefficient of the material are only related to temperature. Set the thermal conductivity and absorption coefficient corresponding to several temperature points. By piecewise linear interpolation, the thermal conductivity and absorption coefficient at different temperatures can be obtained. The temperature response of the measurement point inside the material is obtained through the simulation calculation of the ablation model. The surrogate model can be established, using the thermal conductivity and surface absorption coefficient as input, and the temperature response of the measurement point as the output.

It can be seen from the results of the thermo-gravimetric experiment that the pyrolysis mainly occurs at 300–800℃, and the absorption coefficient changes little during the experiment. Therefore, select the thermo-physical properties as shown in Tab. 2.

Tab. 2 Thermophysical parameter setting

Variable	Parameter
Thermal conductivity	k_{20} k_{100} k_{300} … k_{900} k_{1500}
Absorption coefficient	α

Among Tab. 2, $k_1 - k_{1500}$ are the thermal conductivity corresponding to the temperature in the subscript, and the parameter values of other temperature points can be obtained by piecewise linear interpolation, and α is the surface absorption coefficients. So the input parameter dimension is $D = 8$.

According to [58,59], the thermal conductivity of ablative heat protection materials is in the range of 0 –1 W/(m·K). Without other prior information, it is assumed that it satisfies a uniform distribution in the above range, and the surface absorption coefficient of the material is evenly distributed between 0.5 and 1.0.

In order to investigate the impact of different sample numbers on the accuracy of the surrogate model, the Latin hypercube sampling method is used to extract six sets of samples with the sample numbers of 50, 75, 100, 200, 300, and 400 in the hypercube space of dimension D. The maximum minimum distance method is used to extract 1000 times for different sample numbers to ensure the filling performance of the samples in the space, and the one with the largest minimum distance is selected as the training sample.

The samples in the above hypercube space are transformed into a sample set within the value range of the above parameters through expansion and translation. The sample distribution in the sample sets are shown in Fig. 3 (take $k_1$and εas an example). It can be seen that the obtained samples by maximum minimum Latin hypercube sampling have a good space filling performance. The software of COMSOL Multiphysics is used to establish and solve the ablation model to obtain the temperature response of the internal measuring point. In the calculation, the heating time of the model is set as 90 s, the data collection interval is 1 s, and the distance between the measuring point and the heated surface was 0 mm, 5 mm, 8 mm, 11 mm, 20 mm respectively. Taking each sample set and corresponding temperature response as the training sample sets, a surrogate model for solving the temperature of the measuring point can be established.

5.2 Dimension Reduction of Temperature

Before using the sample data to establish the surrogate model, the dimensionality reduction of the temperature data of the material measurement points in the sample is performed. Taking the training sample of $N = 200$ as an example, the PCA method can be used to achieve dimensionality reduction. The singular values of $\boldsymbol{A}_{\text{centered}}$ is obtained by singular value decomposition of the centered training sample temperature data, namely the diagonal elements and the principal component variance (PCV) of $\boldsymbol{\Lambda}$, as shown in Fig. 4. Among them, the sum of the first 10 variances accounted for 99.7% of the total variance, and the sum of the first 20 variances accounted for about 99.95% of the total variance. Therefore, the first 20 principal components can contain most of the information of the original data, which can be selected to create a surrogate model.

5.3 Surrogate Model Establishment

The Gaussian process model is selected to establish the surrogate model of the ablation model. In order to investigate the prediction effect of different kernel functions in the Gaussian process model, take the training samples of $N = 200$ as an example, use the square exponential kernel function (SE), neural network kernel function

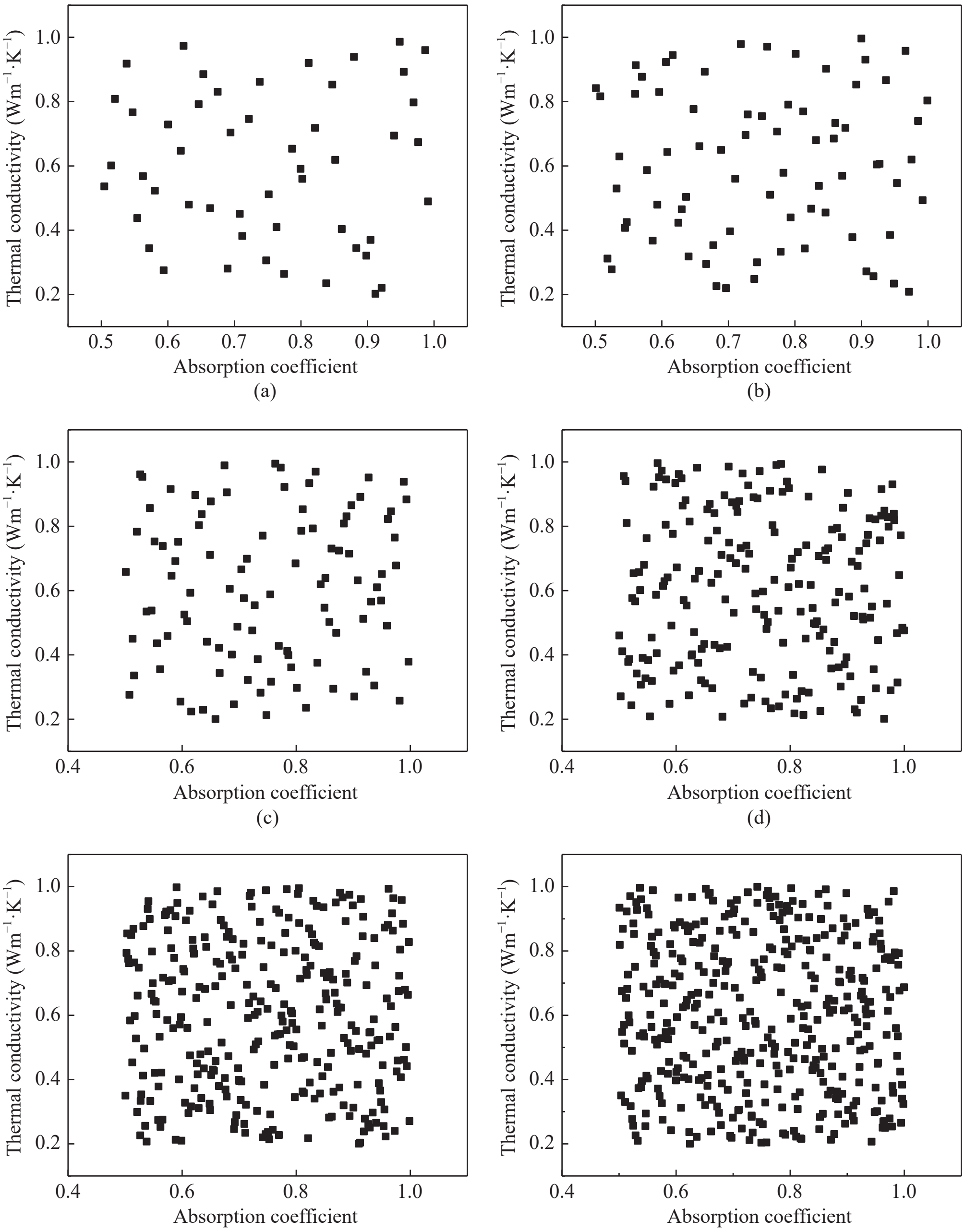


Fig. 3 Sample distribution: (a) 50 samples; (b) 75 samples; (c) 100 samples; (d) 200 samples; (e) 300 samples; (f) 400 samples

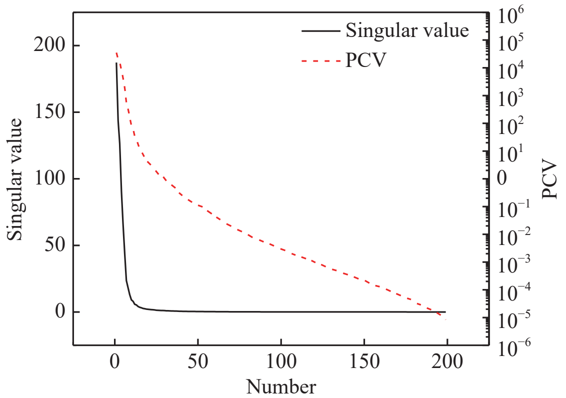


Fig. 4 Singular values of 200 samples

(NN), Matern 3/2 (M32) and Matern 5/2 (M52) four kernel functions respectively to construct surrogate models for the temperature data at different measuring points inside the ablated material at different times. 50 sets of input parameters are randomly selected in the parameter space as a verification sample set to verify the accuracy of the surrogate model, and the ablation model is used for simulation calculation to obtain the temperature history at the measurement point. Use the constructed surrogate model to predict the output corresponding to the parameters of the verification group. The temperature data corresponding to a certain set of thermo-physical parameters are randomly selected, and the temperature prediction results of the surrogate models corresponding to the four kernel functions are compared with the simulation model (SM) results, as shown in Fig. 5. In the figure, "SM" represents the temperature calculated by the simulation models of the three measuring points, and "NN", "M32", "M52", and "SE" respectively represents the temperature history predicted by the surrogate model corresponding to different kernel functions.

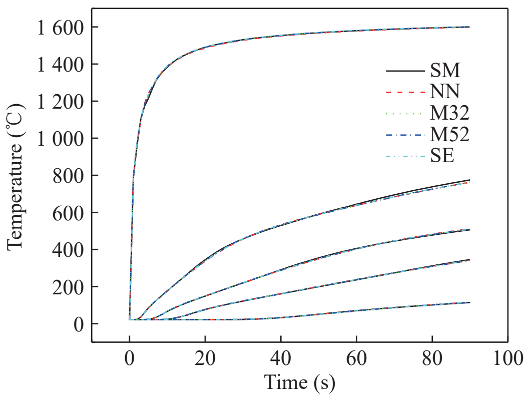


Fig. 5 Comparison of the results of different kernel function surrogate model and numerical model

It can be seen from Fig. 5 that the surrogate models obtained by different kernel functions can predict the temperature changing trend of measuring points, with different prediction errors. The prediction errors between the temperature calculation results of each surrogate model and the numerical model is shown in Fig. 6. For the convenience of observation, only the predicted prediction error of the surface temperature is shown in the figure. It can be seen from the figure that the maximum deviation of the surface temperature prediction is 0– 7 s, which corresponds to the time period when the surface temperature rises quickly, and the maximum deviation is about 2%. The prediction error of the surface temperature is maintained at a low level in other periods, approximately within ±0.25%.

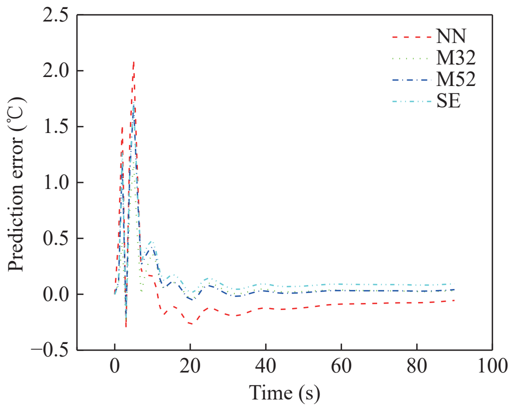


Fig. 6 Deviation of calculation results of surrogate models with different kernel functions

In order to evaluate the effects of different surrogate models, two indicators, the predicted success rate P and the root mean square error (RMSE) of surrogate models, are introduced respectively. The expressions are shown as

$$P = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} S_{ij}}{M * N} \times 100\% \tag{29}$$
$$S_{i,j} = 1 \quad \text{if} \quad |y_{ij} - \hat{y}_{ij}| < B_l$$
$$S_{i,j} = 0 \quad \text{if} \quad |y_{ij} - \hat{y}_{ij}| > B_l$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{M}\sum_{j=1}^{N} (y_{ij} - \hat{y}_{ij})^2}{M * N}} \tag{30}$$

In Eq. (29), S_{ij} indicates whether the predicted value of the ith measuring point at the jth moment of different samples is successful; if it succeeds $S_{ij} = 1$, else $S_{ij} = 0$. B_l indicates the half-width of the confidence interval for the true value. In most engineering models, it usually represents the noise error in the real value. For the calculation data of the ablation model used here, it is set to be 0.2% of the maximum temperature response. For the above four surrogate models, calculate the overall prediction success rate and average root mean square error of the verification sample set, as shown in Tab. 3.

It can be seen from Tab. 3 that among the surrogate models established by the four kernel functions, the surrogate models of the three kernel functions SE, Matern 3/2 and Matern 5/2 perform similarly. Among them, the prediction success rate of SE is the highest, and Matern 5/2 has the smallest root mean square error. The neural network kernel function predicts the lowest success rate, and the root mean square error is also greater.

Tab. 3 Surrogate model indicator calculation results

Indicator	SE	NN	M32S	M52S
P	0.9488	0.8371	0.9410	0.9471
RSME	2.78	5.92	2.88	2.69

6 Optimization of Surrogate Model

In order to reduce the prediction error of the Gaussian process model and improve the prediction success rate, the surrogate model is optimized through the model averaging method and the combined kernel function method.

For the model averaging method, the Bayesian model averaging method is introduced to modify the surrogate models of the three kernel functions of SE, Matern 3/2 and Matern 5/2 to reduce the uncertainty of the surrogate model and improve the prediction accuracy of the model. The model averaging method is a method in which multiple different models are combined with different weight coefficients according to their importance to perform joint prediction. From the prediction results of the surrogate models in the previous section, it can be seen that the prediction accuracy and success rate of the output results of different models are different, and the prediction results of the model can be improved by selecting appropriate weight coefficients. The general expression of the model averaging method is

$$\boldsymbol{a} = \sum_{i=1}^{n} w_i \boldsymbol{a}_i \tag{31}$$

where $\boldsymbol{a}_i$ is the prediction result obtained by using the ith surrogate model; w_i is the corresponding weight coefficient, and n is the number of surrogate models.

Based on the model averaging method, the expression of the Bayesian model averaging method is [60]

$$\boldsymbol{a} = \sum_{i=1}^{n} \boldsymbol{a}_i P(M_i | D) \tag{32}$$

where $P(M_i|D)$ is the probability that the model M_i is the optimal model when the data D is used for model verification.

$$P(M_i|D) = \frac{P(M_i)P(D|M_i)}{\sum_{j=1}^{n} P(M_j)P(D|M_j)} \tag{33}$$

where $P(M_i)$ is the prior probability of M_i being the optimal model. With no other information, it can be considered that each model becomes the optimal model with equal probability. Therefore, Eq. (33) is transformed into

$$P(M_i|D) = \frac{P(D|M_i)}{\sum_{j=1}^{n} P(D|M_j)} \tag{34}$$

where $P(D|M_i)$ is the probability that the data D comes from the model M_i. Assuming that the prediction error of the model M_i is a random variable that satisfies a normal distribution with a mean value of zero and a variance of σ_i^2, then

$$\begin{aligned} P(D|M_i) = & P(d_1, \cdots, d_N | M_i) = \\ & \prod_{k-1}^{N} \left(\left(\frac{1}{2\pi\sigma_i^2} \right)^{1/2} \exp(-1/2) \right) = \\ & \left(\frac{1}{2\pi\sigma_i^2} \right)^{N/2} \exp(-N/2) \end{aligned} \tag{35}$$

For the combined kernel function method, three kernel functions of SE, Mater 32 and Mater 52 are used to construct combined kernel functions:

Combined kernel functions (CKF) 1: $K_1 = K_{SE} + K_{M5/2}$

Combined kernel functions (CKF) 2: $K_1 = K_{SE} + K_{M3/2}$

Combined kernel functions (CKF) 3: $K_1 = K_{M3/2} + K_{M5/2}$

The surrogate models of three combined kernel functions and the weighted surrogate model (WS) obtained by the Bayesian model averaging performed on the surrogate models with three different kernel functions are used to predict the output of the verification sample set. The prediction results and errors are shown in Tab. 4.

Tab. 4 Improved surrogate model indicator calculation results

Indicator	WS	CKF1	CKF2	CKF3
P	0.9554	0.9504	0.9537	0.9501
RSME	2.59	2.68	2.53	2.64

It can be seen from the table that both the weighted surrogate model and the combined kernel function can improve the prediction accuracy of the surrogate model. Compared with the single kernel function, the prediction success rate is improved, and the root mean square error is reduced. The prediction success rate of the weighted surrogate model is slightly higher, because the prediction results are averaged by the weighting coefficients, while the root mean square error of the CKF 2 is relatively smaller. Considering the increase in calculation of different surrogate models caused by the weighted surrogate model, the combined kernel function method of CKF 2 is used when performing parameter identification.

In order to examine the influence of the number of training samples on the accuracy of the surrogate model, the CKF 2 is used as the kernel function, and the surrogate model is established with the training sample sets of 50, 75, 100, 200, 300, and 400 respectively, and the verification data set is used to verify. The success rate and root mean square error of the results are shown in Fig. 7 respectively. It can be seen from the figure that when the sample size enlarges from 50 to 200, the model prediction success rate gradually increases, and the root mean square error gradually becomes smaller. As the sample size continues increasing, the improving the accuracy of the model wanes, indicating that the number of training samples influence less. At this time, the calculation time required to solve the temperature response of the 50 sets of verification sample sets by the surrogate model is about 0.067 s, and the ablation model to solve the temperature response of 1 set of parameters under the same conditions requires about 10 s, which is reduced by more than 99%, and the calculation efficiency is obviously improved.

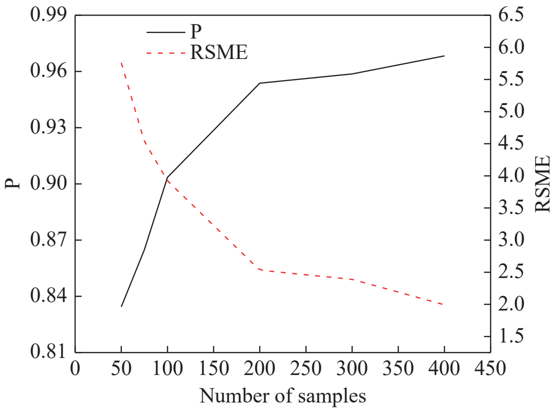


Fig. 7 Prediction results of different sample size models

7 Conclusion

In this paper, Gaussian process model method is used to establish a surrogate model for the thermal response calculation of ablation materials. By comparing the Bayesian model averaging method and the combined kernel function method, the surrogate model is optimized, which improves the model prediction success rate and reduces the root mean square error and computational efficiency increased by 99%.

References:

[1] B. Laub and E. Venkatapathy, “Thermal protection system technology and facility needs for demanding future planetary missions, ” *Planetary Probe Atmospheric Entry and Descent Trajectory Analysis and Science*, 2004, no. 544, pp. 239-247.

[2] Y. Ji, T. Cui, C. Xu, Y. Wang, Z. Li, and Q Luo, “A short overview on low-density ablators for unmanned vehicle, ” in *Proceedings of 2022 International Conference on Autonomous Unman*

[3] D. N. Chandrashekhar, G. G. Raj, and A. G.

Nanjunda, "Properties and application of ablative material, " *International Journal of Engineering Research & Technology*, vol. 10, no. 5, pp. 601-606, 2021.

[4] X Wu, X Lu, T Li, and Z Yi, "A review of researches of light-weight ablators, " *Spacecraft Environment Engineering*, vol. 28, no. 4, pp. 313-317, 2011.

[5] C. Haiming, H. Changqing, and Z. Xinghong, "An overview on low-density ablators, " *Journal of Harbin Institute of Technology*, vol. 50, no. 5, pp. 1-11, 2018.

[6] B. Helber, A. Turchi, J. B. Scoggins, A. Hubin, and T. E. Magin, "Experimental investigation of ablation and pyrolysis processes of carbon-phenolic ablators in atmospheric entry plasmas, " *International Journal of Heat and Mass Transfer*, vol. 100, pp. 810-824, 2016.

[7] C. Ding, X. Liu, F. Xie, C. Du, and Y. Wang, "Heat transfer and pyrolysis gas flow characteristics of light-weight ablative thermal protection system in the blunt body, " *International Journal of Thermal Sciences*, vol. 186, pp. 108-122, 2023.

[8] M. Z. Hasan, "Thermal response and surface recession of a carbon-phenolic charring heatshield of spacecraft: Numerical simulation and validation, " *Journal of Space Safety Engineering*, vol. 9, no. 3, pp. 298-318, 2022.

[9] S. Shan and G. G. Wang, "Survey of modeling and optimization strategies to solve high-dimensional design problems with computationally-expensive black-box functions, " *Structural and Multidisciplinary Optimization*, vol. 41, no. 2, pp. 219-241, 2010.

[10] G. G. Wang and S. Shan, "Review of metamodeling techniques in support of engineering design optimization, " *Journal of Mechanical Design*, vol. 129, no. 4, pp. 1-42, 2006.

[11] R. Jin, W. Chen, and T. W. Simpson, "Comparative studies of metamodelling techniques under multiple modelling criteria, " *Structural and Multidisciplinary Optimization*, vol. 23, no. 1, pp. 1-13, 2001,doi: 10.1007/s00158-001-0160-4.

[12] S. S. Garud, I. A. Karimi, and M. Kraft, "Design of computer experiments: A review, " *Computers and Chemical Engineering*, vol. 106, pp. 71-95, 2017.

[13] F. J. Hickernell, "A generalized discrepancy and quadrature error bound, " *Mathematics of Computation of the American Mathematical Society*, vol. 67, no. 221, pp. 299-322, 1998.

[14] C. Lemieux, *Monte Carlo and Quasi-Monte Carlo sampling and its applications*. New York: Springer Science & Business Media, 2009.

[15] M. E. Johnson, L. M. Moore, and D. Ylvisaker, "Minimax and maximin distance designs, " *Journal of Statistical Planning and Inference,* vol. 26, no. 2, pp. 131-148, 1990.

[16] M. D. McKay, R. J. Beckman, and W. J. Conover, "Comparison of three methods for selecting values of input variables in the analysis of output from a computer code, " *Technometrics*, vol. 21, no. 2, pp. 239-245, 1979.

[17] T. W. Simpson, D. K. J. Lin, and W. Chen, "Sampling strategies for computer experiments: Design and analysis, " *International Journal of Reliability and Applications*, vol. 53, no. 9, pp. 1689-1699, 2013.

[18] J. L. Deutsch and C. V. Deutsch, "Latin hypercube sampling with multidimensional uniformity, " *Journal of Statistical Planning and Inference*, vol. 142, no. 3, pp. 763-772, 2012.

[19] X. L. Liu, Y. W. Chen, X. R. Jing, and Y. G. Chen, "Optimized latin hypercube sampling method and its application, " *Journal of National University of Defense Technology*, vol. 33, no. 5, pp. 73-77, 2011.

[20] H. Zhu, L. Liu, T. Long, and L. Peng, "A novel algorithm of maximin Latin hypercube design using successive local enumeration, " *Engineering Optimization*, vol. 44, no. 5, pp. 551-564, 2012.

[21] D. G. T. Denison, "Simulation based Bayesian nonparametric regression methods, " Ph. D. dissertation, London University, 1997.

[22] V. R. Joseph, Y. Hung, and A. Sudjianto, "Blind kriging: A new method for developing metamodels, " *Journal of Mechanical Design, Transactions of the ASME*, vol. 130, no. 3, pp. 031102, 2008.

[23] G. Aversano, A. Bellemans, Z. Li, A. Coussement, O. Gicquel, and A. Parente, "Application of reduced-order models based on PCA & kriging for the development of digital twins of reacting flow applications, " *Computers and Chemical Engineering*, vol. 121, pp. 422-441, 2019.

[24] J. P. C. Kleijnen, "Kriging metamodeling in simulation kriging metamodeling in simulation: A review, " *European Journal of Operational Research*, vol. 192, no. 3, pp. 707-716 , 2009.

[25] R. G. Regis and C. A. Shoemaker, "Parallel radial basis function methods for the global optimization

of expensive functions, " *European Journal of Operational Research*, vol. 182, no. 2, pp. 514-535, 2007.

[26] R. G. Regis and C. A. Shoemaker, "A stochastic radial basis function method for the global optimization of expensive functions, " *INFORMS Journal on Computing*, vol. 19, no. 4, pp. 497-509, 2007.

[27] M. Kaufman, "Variable-complexity response surface approximations for wing structural weight in HSCT design, " *M. D. thesis*, Virginia Polytechnic Institute and State University, 1996.

[28] B. M. Rutherford, T. L. Paez, L. P. Swiler, and A. Urbina, "Response surface (meta-model) methods and applications, " in *Conference Proceedings of the Society for Experimental Mechanics Series*, 2006.

[29] S. R. Sain and V. N. Vapnik, "The nature of statistical learning theory, " *Technometrics,* vol. 38, no. 4, pp. 409, 1996.

[30] O. Hamidi, J. Poorolajal, M. Sadeghifar, H. Abbasi, Z. Maryanaji, H. R. Faridi, and L. Tapak, "A comparative study of support vector machines and artificial neural networks for predicting precipitation in Iran, " *Theoretical and Applied Climatology*, vol. 119, no. 3-4, pp. 723-731, 2014.

[31] M. Sharifzadeh, A. Sikinioti-Lock, and N. Shah, "Machine-learning methods for integrated renewable power generation: A comparative study of artificial neural networks, support vector regression, and Gaussian process regression, " *Renewable and Sustainable Energy Reviews*, vol. 108, pp. 513-538, 2019.

[32] N. E. Owen, "A comparison of polynomial chaos and Gaussian process emulation for uncertainty quantification in computer experiments, " Ph. D. dissertation, University of Exeter, 2017.

[33] M. M. Rajabi, "Review and comparison of two meta-model-based uncertainty propagation analysis methods in groundwater applications: polynomial chaos expansion and Gaussian process emulation, " *Stochastic Environmental Research and Risk Assessment*, vol. 33, pp. 607-631, 2019.

[34] J. P. C. Kleijnen, "Regression and kriging metamodels with their experimental designs in simulation: A review, " *European Journal of Operational Research*, vol. 256, no. 1, pp. 1-16, 2017.

[35] J. P. C. Kleijnen and E. Mehdad, "Multivariate versus univariate kriging metamodels for multi-response simulation models, " *European Journal of Operational Research*, vol. 236, no. 2, pp. 573-582, 2014.

[36] X. Wu, "Metamodel-based inverse uncertainty quantification of nuclear reactor simulators under the Bayesian framework, " Ph. D. dissertation, University of Illinois at Urbana-Champaign, 2017.

[37] P. M. Mehta, M. Kubicek, E. Minisci, and M. Vasile, "Surrogate model for probabilistic modeling of atmospheric entry for small NEOs, " in *AIAA Space Flight Mechanics Meeting*, 2016, pp. 1-16.

[38] X Chen, "Studies on aerodynamic structural thermal modeling and reduced order modeling of hypersonic vehicles, " Ph. D. dissertation, Beijing Institute of Technology, 2015.

[39] B. K. Bessire, S. A. Lahankar, and T. K. Minton, "Pyrolysis of phenolic impregnated carbon ablator (PICA), " *ACS Applied Materials and Interfaces,* vol. 7, no. 3, pp. 1383-1395, 2015.

[40] H. W. Wong, J. Peck, R. E. Bonomi, J. Assif, F. Panerai, G. Reinisch, J. Lachaud, and N. N. Mansour, "Quantitative determination of species production from phenol-formaldehyde resin pyrolysis, " *Polymer Degradation and Stability*, vol. 112, pp. 122-131, 2015.

[41] R. A. R. Carl and B. Moyer, "Finite difference solution for the in-depth response of charring materials considering surface chemical and energy balances, " NASA CR-1061, 1968.

[42] Y. K. Chen and F. S. Milos, "Ablation and thermal response program for spacecraft heatshield analysis, " *Journal of Spacecraft and Rockets*, vol. 36, no. 3, pp. 475-483, 1999.

[43] A. D. Baer, J. H. Hedges, J. D. Seader, K. M. Jayakar, and L. H. Wojcik, "Polymer pyrolysis over a wide range of heating rates, " *AIAA Journal*, vol. 15, no. 10, pp. 1398-1404, 1977.

[44] Y. K. Chen and F. S. Milos, "Fully implicit ablation and thermal analysis program, " in *Fourth International Conference on Composites Engineering,* New Orleans, LA, pp. 661-662 1997.

[45] H. I. Ene and E. Sanchez-Palencia, "On thermal equation for flow in porous media, " *International Journal of Engineering Science*, vol. 20, no. 5, pp. 623-630, 1982.

[46] Y. Zhu, "Research on the Identification of The Thermal Environment and the Thermal Ablation Response of Thermal Protection Materials, " Ph. D. dissertation, Harbin Institute of Technology, 2018.

[47] S. Ouyang, G. Zheng, and X. Yang, "A theory for the ablation of silica-reinforced phenolic resin plastics and the experimental data, " *Journal of Astronautics*, no. 1, pp. 14-22, 1981.

[48] J. Sacks, W. J. Welch, T. J. Mitchell, and H. P. Wynn, "Design and analysis of computer experiments, " *Statistical Science*, no. 4, pp. 409-435, 1989.

[49] C. E. Rasmussen and C. K. I. Williams, *Gaussian processes for machine learning*. London: MIT Press, 2006.

[50] N. E. I. Gado, E. Grall-Maës, and M. Kharouf, "Linear discriminant analysis based on fast approximate SVD, " in *ICPRAM 2017 - Proceedings of the 6th International Conference on Pattern Recognition Applications and Methods*, vol. 2017, no. 2, pp. 359-365, 2017.

[51] Y. Zhang and Y. Jiang, "Multimodal linear discriminant analysis via structural sparsity, " in *International Joint Conference on Artificial Intelligence*, pp. 3448-3454, 2017.

[52] J. Bai and J. Feng, "Robust principal components analysis with non-sparse errors, " *arXiv: Econometrics,* 2019.

[53] D. García-Gil, S. Ramírez-Gallego, S. García, and F. Herrera, "Principal components analysis random discretization ensemble for big data, " *Knowledge-Based Systems*, vol. 150, pp. 166-174, 2018.

[54] T. Hsing and R. Eubank, *Theoretical foundations of functional data analysis, with an introduction to linear operators*. West Sussex: Wiley, 2013.

[55] A. Buja, D. F. Swayne, M. L. Littman, N. Dean, H. Hofmann, and L. Chen, "Data visualization with multidimensional scaling, " *Journal of Computational and Graphical Statistics*, vol. 17, no. 2, pp. 444-472, 2008.

[56] G. A. O. Hongbin, H. O. U. Jie, and L. I. Ruiguang, "Research on dimension reduction of data stream based on kernel principal com- ponent analysis, " *Computer Engineering and Applications*, vol. 49, no. 11, pp. 105-109, 2013.

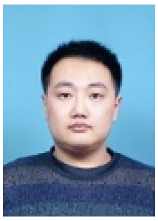
[57] Z. Peng, "Uncertainty analysis and experimental calibration methods of thermal structure model of composite material, " Ph. D. dissertation, Harbin Institute of Technology, 2018.

[58] H. Cheng, H. Xue, C. Hong, and X. Zhang, "Preparation, mechanical, thermal and ablative properties of lightweight needled carbon fibre felt/phenolic resin aerogel composite with a bird's nest structure, " *Composites Science and Technology*, vol. 140, pp. 63-72, 2017.

[59] C. Wang, H. Cheng, C. Hong, X. Zhang, and T. Zeng, "Lightweight chopped carbon fibre reinforced silica-phenolic resin aerogel nanocomposite: Facile preparation, properties and application to thermal protection, " *Composites Part A: Applied Science and Manufacturing*, vol. 112, pp. 81-90, 2018.

[60] Q Yao, "Research and application of metamodel uncertainty quantification based on Bayes' theorem, " M. D. thesis, Huazhong University of Science and Technology, 2015.

Weizhen Pan is a Ph.D. who is currently working in Beijing Institute of Spacecraft System Engineering. At present, the main research directions are the inverse heat transfer problems and the identification of thermal properties of thermal protection material.



Bo Gao is currently an assistant professor at the School of Astronautics, Harbin Institute of Technology. He received his B.S. degree, M.S. degree in Dalian University of Technology, and Ph.D. degree in Harbin institute of technology. The research interests include Quantization of uncertainty, Bayesian methods, Modeling and analysis methods for thermal structure of composite materials and digital twins in spacecraft.